# ST7 - TD 4 Optimal Portfolios: Step-by-Step Solutions

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# 1 Exercise 1: Minimum-Variance Portfolio

## 1. Understanding the Given Data

We are given a market with one riskless asset and two risky assets. The expected returns are given by the vector $R$, and the covariance matrix of the risky assets is $\Sigma$:

$$R = \begin{pmatrix} 0.2 \\ 0.12 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 0.09 & 0.0045 \\ 0.0045 & 0.0225 \end{pmatrix}$$

The riskless asset return is $R^0 = 0.08$. The problem states we are only investing in the risky assets, meaning the weight of the riskless asset is $w_0 = 0$. We also require a "long position" only, meaning our weights must be non-negative ($w_1, w_2 \geq 0$).

Since we must fully invest our capital across the two risky assets, our weights must sum to 1:

$$w_1 + w_2 = 1$$

## 2. Calculating the Portfolio Variance

The variance of a portfolio consisting of two assets, defined by the weight vector $w = (w_1, w_2)^T$, is given by $V(w) = w^T \Sigma w$.

Let's expand this matrix multiplication step-by-step:

$$V(w) = \begin{pmatrix} w_1 & w_2 \end{pmatrix} \begin{pmatrix} 0.09 & 0.0045 \\ 0.0045 & 0.0225 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$$

First, multiply the row vector by the matrix:

$$\begin{pmatrix} w_1 & w_2 \end{pmatrix} \Sigma = \begin{pmatrix} 0.09w_1 + 0.0045w_2 & 0.0045w_1 + 0.0225w_2 \end{pmatrix}$$

Now, multiply this resulting row vector by the column vector $w$:

$$V(w) = (0.09w_1 + 0.0045w_2)w_1 + (0.0045w_1 + 0.0225w_2)w_2$$

$$V(w) = 0.09w_1^2 + 0.0045w_1w_2 + 0.0045w_1w_2 + 0.0225w_2^2$$

Combining the middle terms yields the portfolio variance equation:

$$V(w) = 0.09w_1^2 + 0.009w_1w_2 + 0.0225w_2^2$$

## 3. Formulating the Lagrangian

We want to minimize the variance $V(w)$ subject to the constraint $w_1 + w_2 = 1$, which can be rewritten as $w_1 + w_2 - 1 = 0$.

To do this, we introduce a Lagrange multiplier, $\lambda$, to combine our objective function and our constraint:

$$L(w, \lambda) = V(w) + \lambda(w_1 + w_2 - 1)$$

$$L(w_1, w_2, \lambda) = 0.09w_1^2 + 0.009w_1w_2 + 0.0225w_2^2 + \lambda(w_1 + w_2 - 1)$$

## 4. Solving the First-Order Conditions

To find the minimum, we take the partial derivative of the Lagrangian with respect to $w_1, w_2$, and $\lambda$, and set them equal to zero.

**Derivative with respect to $w_1$:**

$$\frac{\partial L}{\partial w_1} = 2(0.09)w_1 + 0.009w_2 + \lambda = 0.18w_1 + 0.009w_2 + \lambda = 0$$

**Derivative with respect to $w_2$:**

$$\frac{\partial L}{\partial w_2} = 0.009w_1 + 2(0.0225)w_2 + \lambda = 0.009w_1 + 0.045w_2 + \lambda = 0$$

**Derivative with respect to $\lambda$:**

$$\frac{\partial L}{\partial \lambda} = w_1 + w_2 - 1 = 0 \implies w_2 = 1 - w_1$$

We now have a system of linear equations. Since both the $w_1$ and $w_2$ derivatives equal $-\lambda$, we can set them equal to each other:

$$0.18w_1 + 0.009w_2 = 0.009w_1 + 0.045w_2$$

Now, substitute $w_2 = 1 - w_1$ into this equation:

$$0.18w_1 + 0.009(1 - w_1) = 0.009w_1 + 0.045(1 - w_1)$$

$$0.18w_1 + 0.009 - 0.009w_1 = 0.009w_1 + 0.045 - 0.045w_1$$

$$0.171w_1 + 0.009 = -0.036w_1 + 0.045$$

Add $0.036w_1$ to both sides, and subtract 0.009 from both sides:

$$0.207w_1 = 0.036$$

$$\tilde{w}_1 = \frac{0.036}{0.207} \approx 0.1739$$

Using $w_2 = 1 - w_1$, we find:

$$\tilde{w}_2 = 1 - 0.1739 = 0.8261$$

Since both weights are positive ($\tilde{w}_1, \tilde{w}_2 > 0$), this solution naturally satisfies the "long only" condition.

## 5. Expected Value & Variance of the Return

Now we calculate the expected return $r(V)$ and variance $Var(V)$ of this optimized portfolio.
**Expected Return:**

$$r(V) = \tilde{w}^T R = (0.1739)(0.2) + (0.8261)(0.12) \approx 0.03478 + 0.099132 = 0.1339$$

Note that this expected return (13.39%) is greater than the risk-free rate $R^0 = 0.08$.
**Variance:** Using our variance formula $V(w) = 0.09w_1^2 + 0.009w_1w_2 + 0.0225w_2^2$:

$$Var(V) = 0.09(0.1739)^2 + 0.009(0.1739)(0.8261) + 0.0225(0.8261)^2$$

$$Var(V) \approx 0.00272 + 0.00129 + 0.01535 = 0.01936$$

As expected, the minimum variance is strictly positive ($> 0$), meaning uncertainty cannot be completely eliminated.

# 2 Exercise 2: Non-trivial Riskless Portfolio

## 1. Deriving the Expected Returns and Covariance Matrix

We are given the riskless rate $R^0 = 0.06$ and the relationship between the two risky assets:

$$\rho(S^2) = 0.06 + 0.2\rho(S^1)$$

The probability distribution for $\rho(S^1)$ is provided as follows:

| $\rho(S^1)$ & Probability |
|---|
| 0.6 & 0.1 |
| 0.4 & 0.2 |
| 0.3 & 0.4 |
| 0.2 & 0.2 |
| -1 & 0.1 |

**Expected Returns:** First, we calculate the expected return for the first asset, $\mathbb{E}[\rho(S^1)]$:

$$\mathbb{E}[\rho(S^1)] = (0.6)(0.1) + (0.4)(0.2) + (0.3)(0.4) + (0.2)(0.2) + (-1)(0.1)$$

$$\mathbb{E}[\rho(S^1)] = 0.06 + 0.08 + 0.12 + 0.04 - 0.10 = 0.2$$

Using the linear relationship, we can directly find the expected return for the second asset, $\mathbb{E}[\rho(S^2)]$:

$$\mathbb{E}[\rho(S^2)] = 0.06 + 0.2\mathbb{E}[\rho(S^1)] = 0.06 + 0.2(0.2) = 0.06 + 0.04 = 0.1$$

Thus, the expected return vector is $R = \begin{pmatrix} 0.2 \\ 0.1 \end{pmatrix}$.

**Covariance Matrix ($\Sigma$):** Next, we need the variance of $\rho(S^1)$. We first find $\mathbb{E}[\rho(S^1)^2]$:

$$\mathbb{E}[\rho(S^1)^2] = (0.6^2)(0.1) + (0.4^2)(0.2) + (0.3^2)(0.4) + (0.2^2)(0.2) + (-1^2)(0.1)$$

$$\mathbb{E}[\rho(S^1)^2] = 0.036 + 0.032 + 0.036 + 0.008 + 0.10 = 0.212$$

Now, calculate the variance:

$$V(S^1) = \mathbb{E}[\rho(S^1)^2] - (\mathbb{E}[\rho(S^1)])^2 = 0.212 - (0.2)^2 = 0.212 - 0.04 = 0.172$$

Using the properties of variance and covariance with the linear relationship $\rho(S^2) = 0.06 + 0.2\rho(S^1)$:

$$V(S^2) = V(0.06 + 0.2\rho(S^1)) = 0.2^2 V(S^1) = 0.04 \times 0.172 = 0.00688$$

$$Cov(S^1, S^2) = Cov(\rho(S^1), 0.06 + 0.2\rho(S^1)) = 0.2V(S^1) = 0.2 \times 0.172 = 0.0344$$

This gives us the exact covariance matrix provided in the exercise:

$$\Sigma = \begin{pmatrix} 0.172 \& 0.0344 \\ 0.0344 \& 0.00688 \end{pmatrix}$$

Notice that the determinant of $\Sigma$ is zero ($0.172 \times 0.00688 - 0.0344^2 = 0$). The assets are perfectly positively correlated, meaning $\Sigma$ is not invertible and we cannot use the standard matrix optimization from Exercise 1.

## 2. Minimum-Variance Portfolio (Long & Short allowed)

We want to minimize portfolio variance $V(w)$ subject to $w_1 + w_2 = 1$. Let's express the portfolio return $V^w$ strictly in terms of $w_1$ and $\rho(S^1)$.

Substitute $w_2 = 1 - w_1$ and $\rho(S^2) = 0.06 + 0.2\rho(S^1)$ into the portfolio return equation:

$$V^w = w_1\rho(S^1) + w_2\rho(S^2) = w_1\rho(S^1) + (1 - w_1)(0.06 + 0.2\rho(S^1))$$

Expand and factor by $\rho(S^1)$:

$$V^w = w_1\rho(S^1) + 0.06 - 0.06w_1 + 0.2\rho(S^1) - 0.2w_1\rho(S^1)$$

$$V^w = (w_1 + 0.2 - 0.2w_1)\rho(S^1) + 0.06(1 - w_1)$$

$$V^w = (0.2 + 0.8w_1)\rho(S^1) + 0.06(1 - w_1)$$

The variance of this portfolio is simply the variance of the random component:

$$V(w) = Var(V^w) = (0.2 + 0.8w_1)^2 V(S^1) = (0.2 + 0.8w_1)^2(0.172)$$

To find the minimum variance, we can just set this squared term to zero (since variance cannot be negative):

$$0.2 + 0.8w_1 = 0 \implies 0.8w_1 = -0.2 \implies \tilde{w}_1 = -0.25$$

Since $w_1 + w_2 = 1$, we have:

$$\tilde{w}_2 = 1 - (-0.25) = 1.25$$

So the optimal weights are $\tilde{w} = (-0.25, 1.25)^T$. This means we short asset 1 and go long on asset 2.

**Expected Value & Variance:** By construction, the variance of this portfolio is zero:

$$Var(V) = 0$$

The expected return is:

$$r(V) = \tilde{w}^T R = -0.25(0.2) + 1.25(0.1) = -0.05 + 0.125 = 0.075$$

Notice that the risk-free rate is $R^0 = 0.06$, but we have created a portfolio with absolutely no risk ($Var(V) = 0$) that yields $0.075 > 0.06$.

## 3. Creating an Arbitrage Opportunity

Since we found a risk-free portfolio that pays more than the market risk-free rate, we can construct an arbitrage strategy.

Let $\lambda > 0$ be some amount of capital. We construct our strategy at time $t = 0$:

1. Borrow $\lambda$ dollars at the risk-free rate $R^0 = 0.06$. (Cash flow: $+\lambda$)
2. Invest this $\lambda$ into our zero-variance portfolio. Since our weights are $\tilde{w}_1 = -0.25$ and $\tilde{w}_2 = 1.25$:
   - Short sell $0.25\lambda$ worth of asset 1. (Cash flow: $+0.25\lambda$)
   - Buy $1.25\lambda$ worth of asset 2. (Cash flow: $-1.25\lambda$)

Total initial cash flow at $t = 0$ is: $\lambda + 0.25\lambda - 1.25\lambda = 0$. We need zero of our own money to start this operation.

At time $t = 1$, we liquidate the portfolio and pay back the loan:

1. Pay back the borrowed money with interest: $-\lambda(1 + R^0) = -\lambda(1.06)$.
2. Repurchase asset 1 to close the short position: $-0.25\lambda(1 + \rho(S^1))$.
3. Sell asset 2 from our long position: $+1.25\lambda(1 + \rho(S^2))$.

Let's calculate the total payoff at $t = 1$, substituting $\rho(S^2) = 0.06 + 0.2\rho(S^1)$:

$$\text{Payoff} = -\lambda(1.06) - 0.25\lambda(1 + \rho(S^1)) + 1.25\lambda(1 + 0.06 + 0.2\rho(S^1))$$

$$\text{Payoff} = \lambda\left[-1.06 - 0.25 - 0.25\rho(S^1) + 1.25(1.06) + 1.25(0.2)\rho(S^1)\right]$$

$$\text{Payoff} = \lambda\left[-1.31 - 0.25\rho(S^1) + 1.325 + 0.25\rho(S^1)\right]$$

The $\rho(S^1)$ terms perfectly cancel out, mathematically proving our risk is strictly zero.

$$\text{Payoff} = \lambda(-1.31 + 1.325) = 0.015\lambda$$

Because $0.015\lambda$ is strictly positive and guaranteed regardless of what the market does, we have mathematically proven an arbitrage opportunity. The larger the $\lambda$ we borrow, the larger our risk-free profit becomes.

# 3 Exercise 3: Robust Optimization Under Drift Uncertainty

## 1. Understanding the Robust Optimization Problem

In this exercise, we are no longer certain about the expected return vector $R$. Instead, the expected return is a random vector $Z$ with a known expectation $\overline{Z}$ and a positive definite covariance matrix $\Omega$.

Because we are uncertain about the true expected return, we want to optimize our portfolio for the *worst-case scenario* within a reasonable confidence interval. This is represented by the set $G$, which bounds the standardized distance of $Z$ from its mean $\overline{Z}$ by a constant $K$:

$$G := \left\{ z \in \mathbb{R}^2 : (z - \overline{Z})^T \Omega^{-1} (z - \overline{Z}) \leq K^2 \right\}$$

When $K$ is large enough, the true expected return falls within $G$ with high probability.

Our goal is to find the portfolio weights $w \in F$ that maximize our objective, while assuming the market will pick the $z \in G$ that minimizes it. This leads to a max-min (or sup-inf) formulation:

$$\sup_{w \in F} \left( \inf_{z \in G} w^T z - \frac{\lambda}{2} w^T \Sigma w \right)$$

We define the inner minimization problem as the value function $h(w)$:

$$h(w) := \inf_{z \in G} w^T z - \frac{\lambda}{2} w^T \Sigma w$$

The problem is well-posed because $h(w)$ is strictly concave. It is bounded above by $-cw^T \Sigma w$ for some constant $c > 0$, guaranteeing a unique maximum for the outer problem over the feasible set $F$.

## 2. Solving the Inner Minimization for $h(w)$

We need to find the specific vector $z^*$ that minimizes $h(w)$. We set up the Lagrangian for the inner minimization problem. Let $\mu \geq 0$ be the Lagrange multiplier for the inequality constraint:

$$L(z, \mu) = w^T z - \frac{\lambda}{2} w^T \Sigma w + \mu \left( (z - \overline{Z})^T \Omega^{-1} (z - \overline{Z}) - K^2 \right)$$

To simplify the differentiation, let us define a shifted variable $\hat{z} = z - \overline{Z}$. This means $z = \hat{z} + \overline{Z}$. Substituting this into the Lagrangian gives:

$$\hat{L}(\hat{z}, \mu) = w^T (\hat{z} + \overline{Z}) - \frac{\lambda}{2} w^T \Sigma w + \mu (\hat{z}^T \Omega^{-1} \hat{z} - K^2)$$

Now, we take the first-order condition with respect to the vector $\hat{z}$ and set it to zero:

$$\frac{\partial \hat{L}}{\partial \hat{z}} = w + 2\mu \Omega^{-1} \hat{z} = 0$$

Solving for $\hat{z}$, we get:

$$2\mu \Omega^{-1} \hat{z} = -w \implies \hat{z}^* = -\frac{1}{2\mu} \Omega w$$

We know that at the optimum, the constraint binds, meaning $\hat{z}^T \Omega^{-1} \hat{z} = K^2$. Substitute our expression for $\hat{z}^*$ into this constraint to solve for $\mu$:

$$\left( -\frac{1}{2\mu} \Omega w \right)^T \Omega^{-1} \left( -\frac{1}{2\mu} \Omega w \right) = K^2$$

$$\frac{1}{4\mu^2} w^T \Omega^T \Omega^{-1} \Omega w = K^2$$

Since $\Omega$ is a covariance matrix, it is symmetric ($\Omega^T = \Omega$). Thus, $\Omega^T\Omega^{-1}\Omega = \Omega\Omega^{-1}\Omega = \Omega$.

$$\frac{1}{4\mu^2} w^T \Omega w = K^2 \implies \frac{1}{2\mu} = \frac{K}{\sqrt{w^T \Omega w}}$$

Substitute this term back into our expression for $\hat{z}^*$:

$$\hat{z}^* = -\frac{K}{\sqrt{w^T \Omega w}} \Omega w$$

Finally, reverting to our original variable $z^* = \overline{Z} + \hat{z}^*$, we get the worst-case return vector:

$$z^*(w) = \overline{Z} - \frac{K}{\sqrt{w^T \Omega w}} \Omega w$$

## 3. Solving the Outer Maximization for $w^*$

Now we substitute $z^*(w)$ back into our value function $h(w)$:

$$h(w) = w^T \left( \overline{Z} - \frac{K}{\sqrt{w^T \Omega w}} \Omega w \right) - \frac{\lambda}{2} w^T \Sigma w$$

$$h(w) = w^T \overline{Z} - \frac{K}{\sqrt{w^T \Omega w}} w^T \Omega w - \frac{\lambda}{2} w^T \Sigma w = w^T \overline{Z} - K\sqrt{w^T \Omega w} - \frac{\lambda}{2} w^T \Sigma w$$

We want to maximize $h(w)$ subject to the budget constraint $w_1 + w_2 = 1$, which is written in vector notation as $w^T \mathbf{1} - 1 = 0$ (where $\mathbf{1}$ is a vector of ones).

We formulate a new Lagrangian for this outer problem, using $\mu \in \mathbb{R}$ as the multiplier for the budget constraint:

$$\mathcal{L}(w, \mu) = w^T \overline{Z} - K\sqrt{w^T \Omega w} - \frac{\lambda}{2} w^T \Sigma w + \mu(w^T \mathbf{1} - 1)$$

To find the optimal portfolio $w^*$, we take the derivative with respect to the vector $w$:

$$\frac{\partial \mathcal{L}}{\partial w} = \overline{Z} - K \frac{2\Omega w}{2\sqrt{w^T \Omega w}} - \lambda \Sigma w + \mu \mathbf{1} = 0$$

Rearranging terms, we find the system of equations that defines the optimal wealth $w^*$:

$$\lambda \Sigma w^* + \frac{K}{\sqrt{w^{*T} \Omega w^*}} \Omega w^* = \overline{Z} + \mu \mathbf{1}$$

$$w^{*T} \mathbf{1} = 1$$

This non-linear system characterizes the robust optimal portfolio.

## 4. Solving the System for a Specific Case

We are given specific values: $K = 1$, $\Omega = I$ (the identity matrix), and

$$\overline{Z} = \begin{pmatrix} 0.1 \\ 0.1 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 0.01 \& 0.005 \\ 0.005 \& 0.01 \end{pmatrix}$$

Substitute these into our system. Since $\Omega = I$, $\Omega w^* = w^*$ and $w^{*T} \Omega w^* = w^{*T} w^* = |w^*|^2$. The system simplifies to:

$$\lambda \Sigma w^* + \frac{w^*}{|w^*|} = (\mu + 0.1)\mathbf{1}$$

$$w_1^* + w_2^* = 1$$

Notice the symmetry in the given parameters: both assets have the same expected return (0.1), the same variance (0.01), and the same covariance with each other (0.005). Because the optimization problem is

strictly concave and entirely symmetric with respect to both assets, the unique optimal weights must also be symmetric.

Therefore, $w_1^* = w_2^*$. Since they must sum to 1:

$$2w_1^* = 1 \implies \tilde{w} = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$$

# 4 Exercise 4: Portfolio Optimization with Variance Constraint

## 1. Formulating the Lagrangian

In this problem, we want to maximize the expected return of our portfolio, $w^T R$, subject to two constraints: 1. The portfolio variance cannot exceed a given threshold: $w^T \Sigma w \leq \sigma^2$ 2. The weights must sum to 1: $w^T \mathbf{1} = 1$

To solve this constrained optimization problem, we set up the Lagrangian. We introduce a multiplier $\lambda \geq 0$ for the variance constraint and a multiplier $\theta \in \mathbb{R}$ for the budget constraint. Since we are maximizing, we subtract the constraints (or equivalently, add them with specific signs to maintain the correct inequality direction):

$$L(w, \lambda, \theta) = w^T R + \frac{\lambda}{2}(\sigma^2 - w^T \Sigma w) + \theta(w^T \mathbf{1} - 1)$$

Note that the $\frac{1}{2}$ is added for convenience when taking the derivative, as it will cancel out the 2 from differentiating the quadratic form.

## 2. First-Order Conditions (FOC)

We take the partial derivatives of the Lagrangian with respect to $w$, $\lambda$, and $\theta$, and set them to zero.

**Derivative with respect to $w$:**

$$\frac{\partial L}{\partial w} = R - \lambda \Sigma w + \theta \mathbf{1} = 0$$

Solving for $w$, we find the general form of our optimal portfolio $\tilde{w}$:

$$\lambda \Sigma \tilde{w} = R + \theta \mathbf{1} \implies \tilde{w} = \frac{1}{\lambda} \Sigma^{-1}(R + \theta \mathbf{1})$$

**Derivatives with respect to $\lambda$ and $\theta$:** These simply recover our binding constraints:

$$\frac{\partial L}{\partial \lambda} = \frac{1}{2}(\sigma^2 - w^T \Sigma w) = 0 \implies \tilde{w}^T \Sigma \tilde{w} = \sigma^2$$

$$\frac{\partial L}{\partial \theta} = w^T \mathbf{1} - 1 = 0 \implies \tilde{w}^T \mathbf{1} = 1$$

## 3. Solving for the Multipliers

To find the exact values of $\lambda$ and $\theta$, we substitute our expression for $\tilde{w}$ into the two constraints. Let us define three useful scalars to make the notation cleaner:

$$a = \mathbf{1}^T \Sigma^{-1} \mathbf{1}, \quad b = R^T \Sigma^{-1} \mathbf{1}, \quad c = R^T \Sigma^{-1} R$$

**Using the budget constraint:**

$$\tilde{w}^T \mathbf{1} = \left( \frac{1}{\lambda} \Sigma^{-1}(R + \theta \mathbf{1}) \right)^T \mathbf{1} = \frac{1}{\lambda}(R^T + \theta \mathbf{1}^T) \Sigma^{-1} \mathbf{1} = 1$$

$$\frac{1}{\lambda}(R^T \Sigma^{-1} \mathbf{1} + \theta \mathbf{1}^T \Sigma^{-1} \mathbf{1}) = 1$$

Substituting our defined scalars:

$$\frac{1}{\lambda}(b + a\theta) = 1 \implies \lambda = b + a\theta$$

**Using the variance constraint:**

$$\tilde{w}^T \Sigma \tilde{w} = \left( \frac{1}{\lambda} \Sigma^{-1}(R + \theta \mathbf{1}) \right)^T \Sigma \left( \frac{1}{\lambda} \Sigma^{-1}(R + \theta \mathbf{1}) \right) = \sigma^2$$

$$\frac{1}{\lambda^2}(R^T + \theta \mathbf{1}^T)\Sigma^{-1}\Sigma\Sigma^{-1}(R + \theta \mathbf{1}) = \sigma^2$$

Since $\Sigma^{-1}\Sigma = I$ (the identity matrix), this simplifies to:

$$(R^T + \theta \mathbf{1}^T)\Sigma^{-1}(R + \theta \mathbf{1}) = \lambda^2\sigma^2$$

Expanding the product:

$$R^T\Sigma^{-1}R + \theta R^T\Sigma^{-1}\mathbf{1} + \theta \mathbf{1}^T\Sigma^{-1}R + \theta^2\mathbf{1}^T\Sigma^{-1}\mathbf{1} = \lambda^2\sigma^2$$

$$c + 2b\theta + a\theta^2 = \lambda^2\sigma^2$$

Now, substitute $\lambda = b + a\theta$ into this variance equation:

$$c + 2b\theta + a\theta^2 = (b + a\theta)^2\sigma^2$$

$$c + 2b\theta + a\theta^2 = (b^2 + 2ab\theta + a^2\theta^2)\sigma^2$$

Rearranging into a standard quadratic equation in terms of $\theta$:

$$a(a\sigma^2 - 1)\theta^2 + 2b(a\sigma^2 - 1)\theta + (b^2\sigma^2 - c) = 0$$

## 4. Existence and Uniqueness of the Optimizer

To find if an optimal solution exists, we look at the discriminant ($\Delta$) of this quadratic equation:

$$\Delta = [2b(a\sigma^2 - 1)]^2 - 4[a(a\sigma^2 - 1)][b^2\sigma^2 - c]$$

Factoring out $4(a\sigma^2 - 1)$ gives:

$$\Delta = 4(a\sigma^2 - 1)\left[b^2(a\sigma^2 - 1) - a(b^2\sigma^2 - c)\right]$$

$$\Delta = 4(a\sigma^2 - 1)(ab^2\sigma^2 - b^2 - ab^2\sigma^2 + ac)$$

$$\Delta = 4(a\sigma^2 - 1)(ac - b^2)$$

For the optimizer to exist, we need $\Delta \geq 0$. Because $\Sigma$ is a positive definite covariance matrix, the term $(ac - b^2)$ is strictly positive. Therefore, the condition for existence relies entirely on $(a\sigma^2 - 1)$:

$$a\sigma^2 - 1 > 0 \implies \sigma^2 > \frac{1}{a}$$

This makes intuitive sense: $\frac{1}{a}$ is the absolute minimum variance achievable by *any* portfolio on this frontier. If your allowed variance $\sigma^2$ is lower than the global minimum variance, no solution can exist.

Solving for $\theta$ using the quadratic formula yields:

$$\theta^* = \frac{-2b(a\sigma^2 - 1) \pm \sqrt{\Delta}}{2a(a\sigma^2 - 1)} = \frac{-b(a\sigma^2 - 1) \pm \sqrt{(a\sigma^2 - 1)(ac - b^2)}}{a(a\sigma^2 - 1)}$$

Since $\lambda = b + a\theta$, we substitute this back to find $\lambda^*$:

$$\lambda^* = b + a\left(\frac{-b(a\sigma^2 - 1) \pm \sqrt{\Delta/4}}{a(a\sigma^2 - 1)}\right) = \pm\frac{\sqrt{\Delta/4}}{a\sigma^2 - 1}$$

Because we defined our Lagrangian multiplier $\lambda \geq 0$ for an inequality constraint, we must choose the positive root. This guarantees the uniqueness of the solution:

$$\lambda^* = \frac{\sqrt{(a\sigma^2 - 1)(ac - b^2)}}{a\sigma^2 - 1} = \sqrt{\frac{ac - b^2}{a\sigma^2 - 1}}$$

And the unique $\theta^*$ is:

$$\theta^* = \frac{\lambda^* - b}{a} = -\frac{b}{a} + \frac{1}{a}\sqrt{\frac{ac - b^2}{a\sigma^2 - 1}}$$

## 5. Final Expression for the Optimal Weights

Now we substitute our unique $\lambda^*$ and $\theta^*$ back into our initial equation for $\tilde{w}$:

$$\tilde{w} = \frac{1}{\lambda^*}\Sigma^{-1}(R + \theta^*\mathbf{1})$$

$$\tilde{w} = \sqrt{\frac{a\sigma^2 - 1}{ac - b^2}}\Sigma^{-1}\left(R + \left(-\frac{b}{a} + \frac{1}{a}\sqrt{\frac{ac - b^2}{a\sigma^2 - 1}}\right)\mathbf{1}\right)$$

This gives us the exact, closed-form expression for the optimal portfolio weights under a variance constraint.